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SARDAR PATEL UNIVERSITY
B.C.A. (4th SEM NEP) EXAMINATION APRIL - 2026
US04MABCA02 : OPERATING SYSTEMS



Date: 04-04-2026(Saturday)

Time: 02:00 PM to 04:30 PM

NOTE: Right hand figure indicates the marks of each question.

Total Marks: [50]

Q. 1 Multiple Choice Questions (One marks each)

[10]

- 1 A Program in execution is known as _____.
(A) Process (B) Files
(C) Job (D) None of these
- 2 _____ Scheduler select process from ready queue for execution.
(A) Short-term (B) Long-term
(C) Medium-term (D) None of these
- 3 PCB Stands for _____.
(A) Program Control Block (B) Process Control Block
(C) Process Communication Block (D) Pin Communication Block
- 4 _____ is not a valid memory allocation algorithm.
(A) Suit-fit (B) Worst-fit
(C) Next-fit (D) First-fit
- 5 _____ page replacement algorithm suffers from Belady's anomaly.
(A) FIFO (B) Optimal
(C) LRU (D) None of these
- 6 Swapping of process requires _____.
(A) Operating System (B) Main Memory
(C) Backing Store (D) All of these
- 7 _____ is the solution of the bounded buffer.
(A) Shared-Memory (B) Data Consistency
(C) Virtual memory (D) None of these
- 8 Each process has segment of code called _____.
(A) Important Section (B) Critical Section
(C) Mutual Section (D) None
- 9 Which command is used to remove files?
(A) dm (B) rm
(C) delete (D) remove
- 10 For getting present working directory _____ command is used.
(A) wd (B) pwd
(C) pd (D) dir

(1)

P.T.O.

Q. 2 Explain Process Control Block with information stored in it. **[10]**

OR

Q. 2 Explain FCFS and (non-preemptive) Priority Scheduling algorithm with sample process data. **[10]**

Q. 3 Discuss first-fit, best-fit and worst-fit memory allocation algorithms through example and write merits and demerits of each one. **[10]**

OR

Q. 3 Explain Swapping and Paging in detail. **[10]**

Q. 4 Explain Deadlock through appropriate example. Explain all necessary conditions for occurrence of deadlock. **[10]**

OR

Q. 4 Define Co-operative Process with example. Discuss Producer-Consumer Problem in detail. **[10]**

Q. 5 Discuss all conditional statements (Control Structure) using appropriate example. **[10]**

OR

Q. 5 Discuss echo, cd, man, expr and grep command with use, syntax and suitable example. **[10]**

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